WindFuse: Fusing Numerical Weather Prediction and Sparse Observations into a Dense Near-Surface Wind Field over Complex Terrain

Michal KASARDA*[1], Tomáš SVOBODA[1], and Karel ZIMMERMANN[1]

[1]*Department of Cybernetics, Faculty of Electrical Engineering,*

Czech Technical University in Prague, Karlovo náměstí 13, Prague 2 121 35, Czech Republic

ABSTRACT

Near-surface winds over complex terrain remain challenging to predict accurately because kilometre-scale numerical weather prediction (NWP) models smooth out terrain-induced flow features, including ridge acceleration, wake formation, and channelling. Downscaling methods recover finer-scale terrain effects but inherit background forecast errors, while observation-based interpolation is limited by sparse measurements. This paper introduces *WindFuse*, a framework that combines numerical forecasts with local observations to reconstruct wind fields over areas spanning a few kilometres. A neural-network surrogate of computational fluid dynamics (CFD) maps terrain and forecast inflow to a dense, time-averaged, three-dimensional wind field within seconds on an embedded GPU. Exploiting the surrogate's differentiability, WindFuse optimizes two inflow parameters, a wind-speed scaling factor and a wind-direction rotation, using gradient descent to minimize a three-dimensional variational (3D-Var)-type cost function balancing observational misfit against departures from the background forecast, weighted by ensemble spread.

*Corresponding author: Michal KASARDA
Email: kasarmic@cvut.cz

WindFuse is evaluated using synthetic CFD benchmarks, 79 days of observations from the 2017 Perdigão field campaign, and two airborne datasets. At Perdigão, WindFuse achieves the lowest errors among operationally applicable methods using one to eight measurement towers. With a single tower, it reduces the median horizontal wind error from $3.5\,\mathrm{m\,s^{-1}}$ for the background forecast to $2.0\,\mathrm{m\,s^{-1}}$, outperforming downscaling methods (WindNinja and FuXi-CFD) and the observation-based WindSeer baseline, which achieve errors of 2.3 to $2.6\,\mathrm{m\,s^{-1}}$. Applying WindFuse to the inflow fields of FuXi-CFD and WindNinja also improves their performance, demonstrating that its benefits arise from correcting background forecast errors rather than compensating for surrogate-specific biases.

Key words: atmospheric boundary layer, complex terrain, data assimilation, machine-learning surrogate, mountain meteorology, near-surface wind, numerical weather prediction, wind field

Article Highlights:

- WindFuse achieves the lowest error among operationally applicable methods across one to eight towers at Perdigão 2017.
- With one tower, WindFuse reduces median horizontal wind error to $2.0\,\mathrm{m\,s^{-1}}$, outperforming WindNinja, FuXi-CFD and WindSeer.
- Inflow corrections transfer to FuXi-CFD and WindNinja, confirming that WindFuse corrects forecast errors rather than surrogate bias.
- Using CERRA reanalysis instead of TIGGE forecasts, WindFuse maintains similar accu-

racy and outperforms all baselines.

1. Introduction

Operational numerical weather prediction (NWP) models generally capture large-scale atmospheric flow well but struggle to represent near-surface winds over complex terrain. With horizontal grid spacings ranging from approximately one to tens of kilometres, these models cannot resolve the fine-scale topographic features that shape local wind patterns, instead representing the terrain in a smoothed form. Boundary-layer parameterizations, typically developed for relatively homogeneous terrain, often fail to capture terrain-induced phenomena such as flow separation downstream of ridges, speed-up over crests, and channelling through valleys (Serafin et al., 2018). These effects can dominate near-surface wind conditions, with important implications for wind-energy siting, wildfire spread, atmospheric dispersion, and low-altitude aviation. In such applications, terrain-induced winds can even exceed the cruise speed of small uncrewed aerial vehicles (UAVs) (Achermann et al., 2024).

These applications require spatially resolved wind fields across sites spanning a few kilometres, informed by the locally available data. Fire-behaviour models rely on gridded surface winds, which strongly influence fire-spread predictions (Benali et al., 2016). Similarly, hazardous-release dispersion models use wind fields derived from forecasts and nearby observations (Nasstrom et al., 2007), while dense, time-averaged wind fields support safer UAV path planning (Achermann et al., 2019). Across all three applications,

operators typically have access to a background forecast and only a few local observations rather than a dense measurement network. WindFuse is designed specifically for this sparse-observation setting.

Two families of methods address this problem but have complementary limitations (section 2). Downscaling approaches, including nested CFD, mass-consistent solvers, and deep-learning super-resolution, incorporate terrain effects but inherit errors in the forecast inflow (Dujardin and Lehning, 2022). Observation-driven methods can provide accurate estimates near measurement sites but typically require dense observations or have been evaluated primarily over flat terrain (Achermann et al., 2024; Liu et al., 2022). With only a few observations, reconstructing the full wind field is underdetermined. However, forecast errors over scales of a few kilometres are expected to be dominated by coherent biases in wind speed and direction (Jiménez and Dudhia, 2012; Serafin et al., 2018; Xue et al., 2024). A small number of observations may therefore suffice to correct these biases, provided the correction is propagated across the terrain using a flow model rather than simple interpolation.

We propose *WindFuse*, a framework for high-resolution wind estimation that corrects background forecasts using local observations. A neural-network surrogate of computational fluid dynamics (CFD) maps the terrain and forecast inflow to a dense, time-averaged wind field. WindFuse adjusts two inflow parameters, a wind-speed scaling factor and a wind-direction rotation, by minimizing a cost function that balances observational misfit

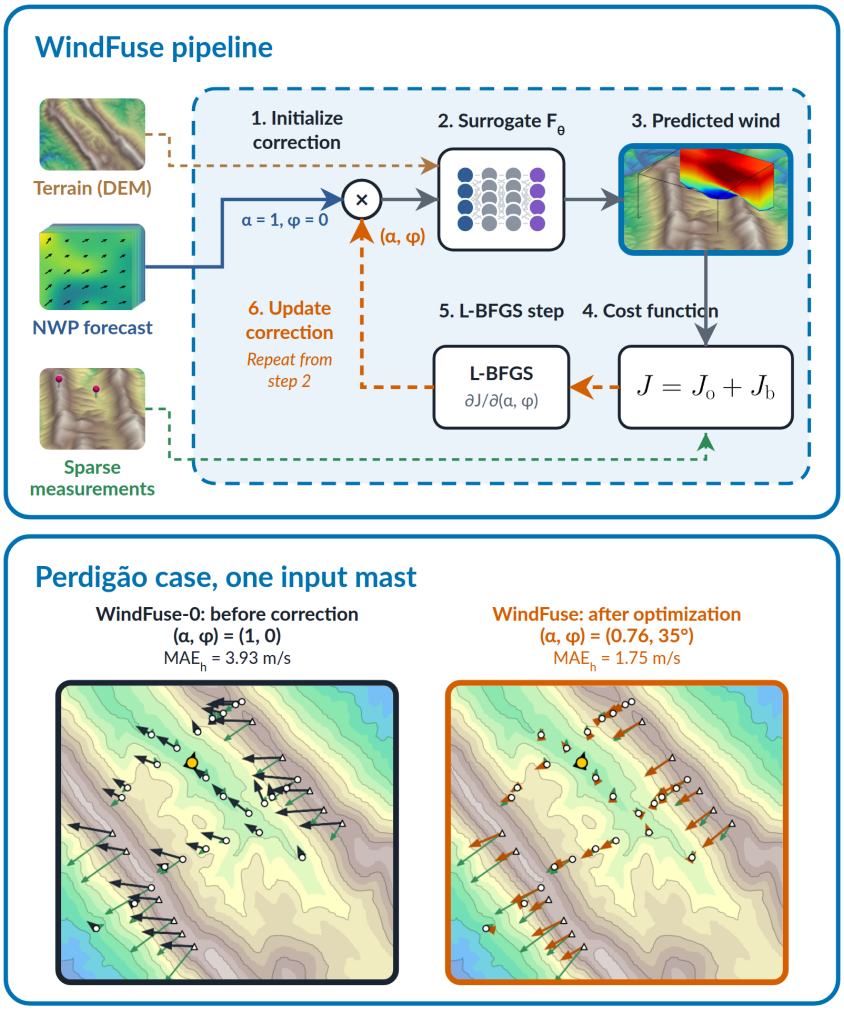


Fig. 1 WindFuse. A two-parameter inflow correction (α, φ), initialized at $(1, 0)$, is applied to the NWP forecast before a frozen CFD surrogate F_θ that also takes the terrain. The predicted dense field is compared with the sparse observations in a cost with an observation term and a background term anchoring (α, φ) to the forecast; L-BFGS updates (α, φ) through the surrogate's gradient. Bottom: a Perdigão case (0700–0800 UTC 21 May 2017), one input tower in the lee (yellow); wind at the verifying towers before and after the correction (coloured arrows) against the observations (grey arrows).

against departures from the background forecast, with each term weighted by its corresponding error model. The forecast thus serves as a prior to be refined rather than a fixed input, allowing the reconstructed field to account for local forecast biases and observations from locations that may poorly constrain the flow (Fig. 1).

This approach formulates wind-field correction as a variational data assimilation problem with a two-parameter control vector and a nonlinear, terrain-resolving observation operator. Operational 3D-Var and 4D-Var methods (Bannister, 2017) minimize similar cost functions at NWP grid scale, typically using linearized observation operators. WindFuse retains the variational framework but replaces the conventional operator with a differentiable neural-network surrogate, enabling efficient gradient-based optimization without explicitly implementing an adjoint. A surrogate evaluation takes milliseconds and a full correction seconds. Whereas operational data assimilation analyses the full model domain using an extensive observing network, WindFuse refines the forecast over an individual site using only a few local observations, complementing rather than replacing operational assimilation.

WindFuse is evaluated on two synthetic CFD benchmarks with controlled forecast errors and three observational datasets: Perdigão 2017, MATERHORN research-aircraft transects, and alpine UAV flights. At Perdigão, it achieves the lowest error among operationally applicable methods using one to eight observation towers, outperforming observation-only (Achermann et al., 2024), forecast-only (Lin et al., 2026), mass-consistent (Wagen-

brenner et al., 2016), and optimal-interpolation baselines. Its inflow correction also improves other downscaling models driven by the same forecast, while requiring only seconds to run on an NVIDIA Jetson.

The remainder of the paper is organized as follows. Section 2 reviews related work; section 3 describes the method and optimization; section 4 presents the data and evaluation setup; section 5 reports the results; section 6 discusses the findings; and section 7 concludes the paper.

2. Related Work

Operational regional models such as ICON (Zängl et al., 2015) and ALADIN (Termonia et al., 2018), together with global ensembles archived in TIGGE (Bougeault et al., 2010), capture synoptic flow but cannot resolve local near-surface wind features at grid spacings of one to tens of kilometres. Statistical post-processing, including model output statistics, analog methods, and machine learning, reduces systematic errors at observing sites over complex terrain (Li et al., 2021; Xue et al., 2024) but does not provide a continuous wind field between stations.

Dynamical and statistical downscaling recover finer-scale flow from gridded forecasts, ranging from microscale CFD within NWP cells (Haupt et al., 2019) to deep-learning super-resolution using GANs, diffusion models, and CNNs (Dujardin and Lehning, 2022; Lian et al., 2024; Lin et al., 2026; Achermann et al., 2019). Mass-consistent models such as WindNinja (Wagenbrenner et al., 2016) are widely used in wildfire management and

dispersion modelling. They enforce mass conservation using domain-average or station-based initialization but lack momentum conservation, limiting their performance on lee slopes (Forthofer et al., 2014). Standard downscaling refines the forecast without local observations, whereas station-initialized WindNinja replaces the forecast rather than combining both sources.

Variational data assimilation, including 3D-Var and 4D-Var (Bannister, 2017), provides a framework for combining forecasts and observations by balancing departures from the background state against observational misfits. Operational systems generally use linearized observation operators and, in 4D-Var, a tangent-linear forecast model, operating at resolutions too coarse to represent terrain-induced flow. Learning-based approaches replace parts of this machinery with differentiable neural networks, including inverse observation operators (Frerix et al., 2021) and reduced-order forward models (Maulik et al., 2022), linking data assimilation and machine learning within a common inverse-problem framework (Geer, 2021). However, these methods remain focused on NWP-scale analysis. At finer scales, Zanetta et al. (2026) combine an NWP prior, topographic descriptors, and station observations in a Gaussian-process neural-network regressor to produce calibrated two-dimensional surface winds on a 250 m grid. This approach statistically blends observations into a wind map without explicitly modelling terrain-resolved flow.

Another research direction reconstructs dense wind fields directly from sparse observations and terrain, without a forecast prior. WindSeer (Achermann et al., 2024) uses a

convolutional 3D encoder–decoder trained on Reynolds-averaged Navier–Stokes (RANS) CFD to reconstruct volumetric winds from terrain and sparse UAV-track or tower measurements in real time on embedded hardware. Other approaches predict urban flow from lidar and sparse observations (Folk et al., 2024), reconstruct mountain winds at multiple elevations using graph neural networks (Gao et al., 2024), or estimate surface wind speed from observations and background fields using random forests trained on long-term urban measurements (Liu et al., 2022). However, observation-driven methods generally lack a synoptic prior to constrain the solution when measurements are sparse or poorly located. The method of Liu et al. (2022) additionally relies on a long record from a dense network and does not incorporate a flow model.

A related body of work estimates flow-model inflow conditions from observations. Wind-resource studies fit CFD inflows to direction-binned tower climatologies (Radünz et al., 2020), while urban applications estimate inflow parameters for polynomial-chaos surrogates using ensemble Kalman filtering (Sousa et al., 2018); neither incorporates a numerical forecast as a prior. Differentiable solvers have also been used to infer boundary conditions through gradient-based optimization in non-atmospheric settings (Fan et al., 2026), but not for real-time, hectometre-scale wind reconstruction over complex terrain. The concurrent WindINR approach (Xiao et al., 2026) updates a low-dimensional latent state of a frozen neural wind model using sparse observations and a Gaussian prior initialized from a kilometre-scale background. However, its correction lacks a direct physical

interpretation and has been evaluated only with synthetic observations over a single terrain.

To our knowledge, no previous study has used real in situ observations to optimize a numerical-weather-prediction inflow correction through a differentiable neural-network CFD surrogate, producing a dense three-dimensional wind field at tens-of-metres resolution over complex terrain.

3. Methodology

We estimate a dense time-averaged wind field by correcting an NWP forecast to agree with sparse in situ observations. Over complex terrain under synoptic forcing, the local wind structure is dominated by the interaction of the synoptic flow with the terrain (Fernando et al., 2019). The forecast captures the synoptic flow, and at the scale of a single site its error is expected to be largely a spatially coherent bias in speed and direction (Jiménez and Dudhia, 2012). We therefore correct the forecast with one scale and one rotation and let a differentiable CFD surrogate propagate the correction through the terrain. Thermally driven flows (Fernando et al., 2015) are deliberately left outside the correction; sections 5.4 and 5.7 show where this matters.

3.1 Problem Formulation

Let T denote the terrain over the area of interest and $\mathscr{N}$ the NWP forecast over the same domain, at any resolution. We are given N observation positions $\mathscr{P} = \{\mathbf{p}_i\}_{i=1}^{N}$ with time-averaged observed wind vectors $\mathscr{Y} = \{\mathbf{y}_i\}_{i=1}^{N}$ (pointing in the direction of flow) and

a per-observation error model (σ_i^s, σ_i^d) for speed and direction. The goal is to recover a dense time-averaged wind field $\mathbf{U}$ over the domain that agrees with the observations while staying consistent with the forecast. A differentiable CFD surrogate F_θ maps the forecast and the terrain to the dense field,

$$\mathbf{U} = F_\theta(\mathscr{N}, T). \tag{1}$$

We require only that F_θ be differentiable with respect to the forecast input.

3.2 Speed–direction correction

Because the NWP error is assumed spatially coherent, we correct the forecast with two interpretable scalars: a multiplicative wind-speed scaling factor $\alpha \in \mathbb{R}_+$ and a wind-direction rotation $\varphi \in [-\pi, \pi)$, positive counter-clockwise, applied to the forecast. Writing $\mathscr{N}_{\alpha,\varphi}$ for the forecast with every horizontal wind vector scaled by α and rotated by $R(\varphi)$, the corrected dense field is

$$\mathbf{U}(\alpha, \varphi) = F_\theta(\mathscr{N}_{\alpha,\varphi}, T). \tag{2}$$

Restricting the correction to these two degrees of freedom matches the hypothesized error structure and keeps the corrected forecast physical: the profile is scaled and rotated as a whole, so the correction cannot chase the observations by reshaping it; higher-order effects are left to the surrogate.

3.3 Cost function

We estimate the correction (α, φ) from the observed wind vectors $\mathscr{Y}$. For each observation we form its model-side counterpart $\hat{\mathbf{y}}_i(\alpha,\varphi) = F_\theta\big(\mathscr{N}_{\alpha,\varphi}, T\big)(\mathbf{p}_i)$. This is the corrected dense field of Eq. (2), sampled at the observation position $\mathbf{p}_i$ by trilinear interpolation, so $\hat{\mathbf{y}}_i$ stays differentiable in (α, φ). We minimize a two-term cost function of 3D-Var form over $\alpha \in \mathbb{R}_+$ and $\varphi \in [-\pi, \pi)$,

$$J(\alpha,\varphi) = J_{\mathrm{o}}(\alpha,\varphi) + J_{\mathrm{b}}(\alpha,\varphi). \tag{3}$$

The observation term J_{o} penalizes the misfit to the observations. The background term J_{b} anchors the correction to the forecast. Both are squared deviations normalized by an assumed Gaussian error variance. We average the observation term over the N observations (Eq. (5)), so the weight of the data relative to the forecast is fixed by the error scales and does not grow with N; the same settings apply to one observation and to a tower array. The average is deliberate: the representation errors of towers on one site are strongly correlated, so summing N terms as if they were independent would overstate the information the observations carry. The average is equivalent to inflating the observation-error variance by N, the standard remedy for correlated representation error (Janjić et al., 2018).

The residual of each observation is $\mathbf{r}_i(\alpha,\varphi) = \hat{\mathbf{y}}_i(\alpha,\varphi) - \mathbf{y}_i$. It is weighted by a standard deviation that combines a speed and a direction component,

$$\sigma_i^2 = (\sigma_i^s)^2 + \big(\|\mathbf{y}_i\| \sin \sigma_i^d\big)^2, \tag{4}$$

where σ_i^d is in radians and $\sin\sigma_i^d$ converts the direction uncertainty into a cross-wind error. We use one effective pair $(\sigma_i^s, \sigma_i^d) = (0.3\,\mathrm{m\,s^{-1}}, 5^\circ)$ for all towers. These are representation-error values: they are dominated by the mismatch between a point sonic and a surrogate grid cell, not by instrument precision; the factor $\|\mathbf{y}_i\|$ makes the cross-wind error grow with wind speed. The observation term is the mean normalized squared residual,

$$J_\mathrm{o}(\alpha,\varphi) = \frac{1}{N}\sum_{i=1}^{N}\frac{\|\mathbf{r}_i\|^2}{\sigma_i^2}. \tag{5}$$

The background term penalizes departures from the uncorrected forecast $(\alpha,\varphi) = (1,0)$,

$$J_\mathrm{b}(\alpha,\varphi) = \left(\frac{\alpha-1}{\sigma_\alpha^\mathrm{eff}}\right)^2 + \left(\frac{\varphi}{\sigma_\varphi^\mathrm{eff}}\right)^2, \tag{6}$$

where $\sigma_\alpha^\mathrm{eff}$ is a relative speed uncertainty and $\sigma_\varphi^\mathrm{eff}$ a rotation uncertainty.

3.4 Background-error specification from the ensemble spread

The background-error standard deviation combines a small base value $(\sigma_\alpha, \sigma_\varphi)$ with the spread $(\sigma_\mathrm{nwp}^u, \sigma_\mathrm{nwp}^v)$ of an ensemble forecast, when one is available. The base value covers effects such as the mismatch between the NWP grid and the local domain; the spread makes the background error flow-dependent. Propagating the spread to first order through the forecast wind speed and direction, at the forecast levels inside the domain, gives

$$\delta_s = \frac{\left\|(u\,\sigma_\mathrm{nwp}^u,\, v\,\sigma_\mathrm{nwp}^v)\right\|}{\|(u,v)\|}, \qquad \delta_\varphi = \frac{\left\|(v\,\sigma_\mathrm{nwp}^u,\, u\,\sigma_\mathrm{nwp}^v)\right\|}{\|(u,v)\|^2}. \tag{7}$$

These are added in quadrature to the base values,

$$(\sigma_\alpha^\mathrm{eff})^2 = \sigma_\alpha^2 + \left(\frac{\langle\delta_s\rangle}{\langle\|(u,v)\|\rangle}\right)^2, \qquad (\sigma_\varphi^\mathrm{eff})^2 = \sigma_\varphi^2 + \langle\delta_\varphi\rangle^2, \tag{8}$$

where $\langle\cdot\rangle$ is the mean over the in-domain forecast levels. The speed term is divided by the mean speed to match the relative σ_α. Without an ensemble, σ^{eff} equals the base value.

3.5 Relation to variational assimilation

In the 3D-Var framework, the control vector is (α, φ), with background $(1, 0)$ and background error covariance $\text{diag}\big((\sigma_\alpha^{\text{eff}})^2, (\sigma_\varphi^{\text{eff}})^2\big)$. The nonlinear observation operator is $\mathscr{H}(\alpha, \varphi) = F_\theta(\mathscr{N}_{\alpha,\varphi}, T)(\mathbf{p}_i)$, and the observation error covariance is $\text{diag}(\sigma_i^2)$. WindFuse differs from operational 3D-Var in three respects: its control vector has only two elements, its observation term is averaged rather than summed, and its observation operator is a differentiable neural network that resolves terrain-induced flow. The operator's gradient replaces the conventional adjoint and can be evaluated in milliseconds.

3.6 Optimization

The network weights remain fixed; only (α, φ) is optimized for each observation set. The fully convolutional surrogate generalizes beyond its training subdomain (section 4) and propagates gradients to both parameters. We minimize Eq. (3) using L-BFGS, as in operational 4D-Var (Bannister, 2017), with a strong-Wolfe line search, step size 0.1, and history size 100. Optimization starts from the background $(\alpha, \varphi) = (1, 0)$ and allows four outer steps, each with up to four inner iterations. The inner loop terminates early when standard gradient or cost-change tolerances are met. The scale factor is constrained to remain above 0.05; if the cost becomes non-finite, the step is reverted and optimization

stops at the last finite iterate.

4. Data and experimental design

4.1 CFD surrogate and training data

We instantiate F_θ with *WindFuse-0*, a 3D CNN based on the WindSeer architecture (Achermann et al., 2024), trained from scratch on the original WindSeer CFD dataset using its original data split. The dataset comprises steady RANS simulations of neutrally stratified flow over real terrain (Achermann et al., 2024). Unlike WindSeer, which takes sparse observations as input, WindFuse-0 takes an inflow wind field. Its inputs are the distance to the terrain and the two horizontal inflow components, while its outputs are the three mean wind components and turbulent kinetic energy.

As the dataset contains no forecast fields, we prescribe a neutral logarithmic inflow profile:

$$U(z) = U_{200}\,\frac{\ln(z/z_0)}{\ln(200\,\mathrm{m}/z_0)}, \qquad z_0 = 0.1\,\mathrm{m}, \tag{9}$$

where U_{200} is determined from the speed and direction of the topmost inlet layer in each CFD sample. The resulting inflow is horizontally uniform, set to zero inside the terrain, and left unperturbed during training.

We train on $64\times64\times64$-grid-point subdomains with horizontal and vertical spacings of 16.5 m and 11.5 m, respectively. Training minimizes the mean squared error across all four outputs over 500 epochs, using Adam with a learning rate of 10^{-4} and exponential decay

and a batch size of 64. On withheld test terrains, the median wind-vector error is 0.79 m s^{-1}, corresponding to 7% of the local wind speed. The WindSeer baseline in section 4.6 uses the same architecture and training data. As its pretrained weights are unavailable, we retrain it using the original setup.

4.2 Forecast backgrounds and inflow

The primary background is the 50-member ECMWF ensemble from the TIGGE archive (Bougeault et al., 2010), provided on a regular 0.5° grid ($\approx$ 50 km; the 2017 ensemble ran at $\approx$ 18 km). At this resolution, each study site is covered by a single grid column. We use the 0000 UTC run for each case day, with forecast steps from 0 to 36 h at 6-h intervals, interpolated linearly in time. This temporal sampling under-resolves the diurnal cycle, and the maximum forecast lead time is 24 h. The ensemble mean defines the inflow, while the ensemble spread determines the prior uncertainty (Eq. (8)).

TIGGE provides winds on eight pressure levels (1000–200 hPa). At Perdigão, the 1000 hPa level (approximately 60 m above sea level) lies below the valley floor, requiring extrapolation below the model orography, whereas the 925 hPa level is approximately 720 m. These two levels bracket the site, so the near-surface inflow is interpolated between them. We use forecasts rather than reanalyses to reflect operational availability. We also select the openly archived TIGGE ensemble rather than a proprietary, higher-resolution forecast because WindFuse is designed to be independent of the choice of background (section 5.5).

For sensitivity testing, we additionally use the CERRA regional reanalysis (Ridal et al., 2024), which has a 5.5 km grid spacing. Winds from the same eight pressure levels are taken from its 6-hourly analyses and interpolated in time as for TIGGE. Because CERRA is deterministic, ensemble-based prior inflation is inactive (section 4.7).

At inference, the forecast replaces the CFD-derived inflow. Each pressure level is assigned its geopotential height. Forecast winds are interpolated bilinearly across the domain and linearly in height between levels, with linear extrapolation beyond the outermost levels. Winds are set to zero inside the terrain. The parameters α and φ uniformly scale and rotate the horizontal wind at all levels.

The archived 10-m wind is excluded from the inflow because it is a diagnosed quantity that already accounts for the modelled surface drag. The 10-m wind is used only to initialize WindNinja, which requires this input (section 4.6).

The surrogate was trained using logarithmic inflow profiles (Eq. (9)), whereas the forecast profile does not follow a logarithmic law near the surface. We therefore apply a fixed, partial log-law taper below the forecast boundary-layer height to the inflow supplied to F_θ, bringing it closer to the training profile. This adjustment adapts the input specifically for the surrogate rather than correcting the forecast itself. It reduces wind speed by at most 25%, introduces no fitted parameters, and is not applied to the raw-forecast baseline (section 4.6; details in the ESM).

Even with an otherwise accurate forecast, we expect $\alpha < 1$ for two reasons. First, the

lowest archived pressure levels lie in the upper boundary layer or above it, where surface friction has had less influence on wind speed. Second, mesoscale models tend to overestimate near-surface winds over complex terrain because unresolved orographic drag is under-represented (Jiménez and Dudhia, 2012; Xue et al., 2024). The fitted α therefore accounts for both this representational bias and forecast error; we do not attempt to separate their contributions.

4.3 Perdigão observations

The Perdigão 2017 campaign (Fernando et al., 2019) investigated a double-ridge site in central Portugal, where two parallel NW–SE ridges lie 1.4 km apart and rise approximately 200 m above the intervening valley (Menke et al., 2019). Prevailing winds cross the ridges from the SW and NE, and lee-side recirculation occurs in more than half of cross-ridge cases (Menke et al., 2019).

Sonic anemometers were deployed on 49 towers across the ridges and valley. Of these, 34 lie within our domain and provide input and verification observations at 14 heights between 2 and 100 m above ground. The lowest measurement levels fall within the surrogate's first 11-m vertical layer. We use the quality-controlled, tilt-corrected 5-min sonic product from the NCAR Earth Observing Laboratory Integrated Surface Flux System (ISFS). A 5-min bin is retained if its diagnostic flag is clear and at least 75% of its 20-Hz samples are available. We average u, v, and w over one bin for 5-min cases and over twelve bins for 1-h cases.

Our evaluation comprises 200 cases spanning 79 days between 11 March and 16 June 2017. These include 99 strong-wind and 101 weak-wind cases, classified using a mean tower-wind threshold of $8\,\mathrm{m\,s^{-1}}$. The cases cover the diurnal cycle, with between 9 and 34 towers reporting valid data per case (median 27). Terrain elevation is taken from the Copernicus GLO-30 digital elevation model (1 arc second, approximately 30 m resolution) (European Space Agency, 2021) and resampled to the surrogate grid. WindFuse-0 and WindSeer use a $128\times128\times64$-grid-point domain, with horizontal and vertical spacings of 16.2 m and 11 m, respectively. FuXi-CFD runs on its native 9-km domain centred on the valley.

For each case, we randomly draw 30 subsets of $N \in \{1,2,4,8\}$ input towers from those available and provide the same subsets to every method. The remaining towers are reserved for verification. This setup targets the sparse-observation regime, with eight input towers representing a denser network. Only OI* uses the full network to estimate its covariances.

4.4 Airborne observations

We use two airborne datasets. The first comes from the MATERHORN fall campaign (Fernando et al., 2015), during which a Twin Otter research aircraft flew transects a few hundred metres above Granite Mountain, Utah. The 23 straight passes, conducted over five days during quiescent intensive observation periods, each define one case. During these periods, the 700-hPa wind was below $5\,\mathrm{m\,s^{-1}}$, and afternoon and evening-transition flows over the isolated mountain were weak and thermally driven, with tower winds of

1–5 m s^{-1}. We fit the correction using 1-Hz flight-level winds and verify it against 5-min sonic averages at ground towers. The domain contains 128^3 grid points at 16.2 m spacing, and the taper reference height is fixed at 1000 m.

The second dataset comes from the WindSeer flight campaign (Achermann et al., 2024), in which clusters of two or three small fixed-wing UAVs flew simultaneously at three Swiss mountain sites: Chasseral in the Jura (1548 m) and the Gotthard (1960 m) and Oberalp (2069 m) passes in the Alps. UAVs within each cluster flew up to 800 m apart. We use the published Kalman-filter reconstruction of wind along each flight track (Achermann et al., 2024). These estimates are noisier than sonic measurements, and their accuracy has not been independently quantified. We divide the flights into synchronized 30-s windows (131 in total), which are substantially shorter than the averaging period of the RANS training fields. We perform leave-one-out evaluation within each cluster: each method receives the window-mean wind from one UAV and is evaluated against a withheld UAV. The surrogates use 64^3-grid-point domains with horizontal and vertical spacings of 16.2 m and 11 m, respectively. As at Perdigão, TIGGE provides the background.

4.5 Synthetic benchmarks

To evaluate the corrected field across the full three-dimensional grid, we construct two synthetic benchmarks with dense CFD truth and prescribed forecast errors (see the ESM for details). *FuXi-CFD-zones* comprises 40 RANS domains drawn from the FuXi-CFD dataset (Lin et al., 2026). *WindSeer-zones* comprises training-size subdomains (section 4.1)

centred on lee and recirculation zones in test-split WindSeer terrains (Achermann et al., 2024).

Each domain contains simulated UAV observations incorporating sensor noise, bias, and gusts, together with forecast-like forcing reconstructed from the CFD fields. In each trial, the forcing is scaled and rotated by a randomly sampled $(\alpha_{\mathrm{err}}, \varphi_{\mathrm{err}})$, with one of three error severities. The correction is supplied with the generating noise level as its prior uncertainty, representing a best-case estimate. Because the imposed errors lie within the correction's parameter space, these benchmarks test identifiability under a matched error model. Structural forecast errors are instead evaluated using real observations.

All methods are evaluated against the CFD truth on the native grid within each evaluation subdomain ($\sim 10^5$ points). FuXi-CFD is evaluated only on FuXi-CFD-zones, which may overlap its training distribution because no dataset split is published. The WindSeer-zones domains are too small for FuXi-CFD's 9-km domain.

4.6 Baselines

Raw NWP interpolates the forecast to the verification points without using terrain or observations. On the synthetic benchmarks, it uses the prescribed forcing; on the real campaigns, it uses the TIGGE ensemble-mean winds interpolated from the pressure levels. This is the same free-atmosphere inflow supplied to the surrogate, without the taper described in section 4.2. The archived 10-m wind is used only by WindNinja; its error against the 10-m sonic measurements is reported in ESM Table S2.

NWP-opt fits the same (α, φ) as WindFuse but applies the correction directly to the raw forecast, without the surrogate, thereby isolating the surrogate's contribution. **WindFuse-0** applies the surrogate to the uncorrected forecast, isolating the benefit of the optimization. **WindSeer** (Achermann et al., 2024) reconstructs the dense wind field from terrain and sparse observations alone, without a forecast. **FuXi-CFD** (Lin et al., 2026) downscales the uncorrected forecast using a pretrained CFD surrogate over a fixed 9×9 km domain at 30 m resolution.

OI combines observations with the forecast background using optimal interpolation, following the classical data-assimilation scheme (Kalnay, 2003). It uses a distance-only Matérn-3/2 covariance with a 500 m length scale and the error variances defined in section 3; the full configuration is provided in the ESM.

OI* is an in-sample variant that uses separate horizontal and vertical length scales (2.2 km and 140 m above sea level, respectively), fitted to forecast–tower innovations across all 200 Perdigão cases. It corresponds to the elevation-dependent OI used in operational surface analyses (Uboldi et al., 2008; Lussana et al., 2018) and provides an upper bound for this class of method at the site, rather than an operational baseline.

WindNinja (Wagenbrenner et al., 2016) is a widely used mass-consistent model for surface winds over complex terrain. We run it at 30 m resolution on the same terrain, initialized with the forecast 10-m wind. Surface roughness and canopy height are derived from land-cover (Zanaga et al., 2022) and canopy-height datasets (Lang et al., 2023), re-

spectively. Its non-neutral stability option is driven by the forecast 2-m temperature and cloud cover. The diurnal slope-wind model is disabled; a sensitivity test with it enabled is reported in the ESM.

4.7 Verification metrics and significance

Each trial is evaluated at its withheld points $\mathscr{E}$. The horizontal vector error is $\mathbf{e}_j = \bar{\mathbf{u}}^{\mathrm{pred}}_{\mathrm{h},j} - \bar{\mathbf{u}}^{\mathrm{obs}}_{\mathrm{h},j}$. The horizontal mean absolute error ($\mathrm{MAE_h}$) and root-mean-square error ($\mathrm{RMSE_h}$) are defined as the mean and root mean square, respectively, of $\|\mathbf{e}_j\|_2$ over $\mathscr{E}$. Speed bias is the mean difference between predicted and observed horizontal wind speed, with positive values indicating overestimation. The horizontal Pearson correlation coefficient ($\mathrm{PCC_h}$) is the mean of the Pearson correlations for the *u* and *v* components over $\mathscr{E}$; it measures agreement in the spatial pattern independently of a constant offset. For airborne cases, we report the corresponding three-dimensional metrics: relative mean absolute error ($\mathrm{RMAE_{3D}}$), defined using the 3D vector error relative to the observed speed floored at $2\,\mathrm{m\,s^{-1}}$, and $\mathrm{PCC_{3D}}$. Box plots show the quartiles, with whiskers extending to 1.5 times the interquartile range.

We report case-level medians with 95% bootstrap confidence intervals, following standard practice in forecast verification (Wilks, 2019). The 200 Perdigão cases span 79 days, and cases from the same day share weather conditions. We therefore resample whole days rather than individual cases, using 2000 draws with replacement. The 2.5th and 97.5th percentiles define the interval bounds. A difference between two methods is considered

significant at the 95% level when the confidence interval for the paired statistic excludes the null value. Paired Wilcoxon signed-rank tests on the 200 per-case values provide a complementary assessment, although they do not account for day-level clustering.

All experiments use the L-BFGS settings described in section 3.6. For the real-data campaigns, the base prior uncertainties are σ_α=0.1 and σ_φ=15°, inflated according to the ensemble spread (section 3.4). These base values account for NWP grid misalignment, uncertainty in the GFS boundary-layer height, and forecast errors not captured by the ensemble spread. For the deterministic CERRA background, ensemble-based inflation is inactive, so the base prior is widened to approximate the typical uncertainty reached after inflation with TIGGE: σ_α=0.2 and σ_φ=20°. The observation errors assumed in the cost function (Eq. (4)) are $0.3\,\mathrm{m\,s^{-1}}$ in wind speed and 5° in direction. These values also account for flow features not represented by the surrogate, such as thermal drafts.

4.8 Use of AI tools

The large language models Claude Opus 5.5 and Claude Fable 5.1 (Anthropic) were used as a coding assistant and for language editing of the manuscript; they did not design the method, the experiments, or the interpretation. The authors checked all code, results, and text and take full responsibility for them.

5. Results

All test domains below lie outside the training set of WindFuse-0 and WindSeer.

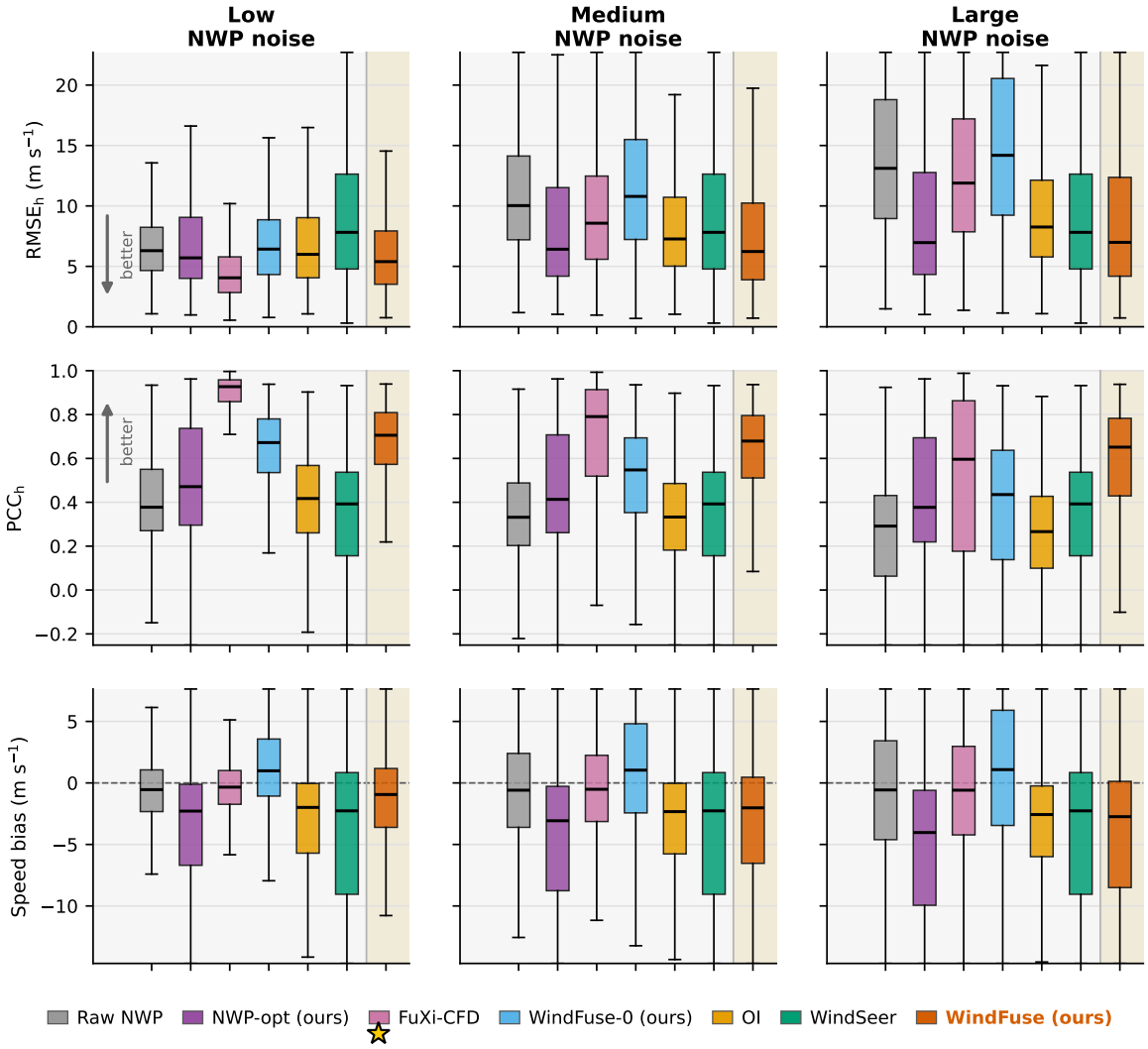


Fig. 2 Synthetic benchmarks, WindSeer-zones and FuXi-CFD-zones pooled, by forecast-corruption severity (2028 FuXi-CFD-zones and 2000 WindSeer-zones trials per corruption level; the low column pools two levels): per-trial horizontal vector RMSE, PCC_h, and speed bias against the dense CFD field. FuXi-CFD ($\star$) runs on FuXi-CFD-zones only.

5.1 Synthetic benchmarks

As the corruption grows, the forecast-driven methods degrade strongly (Fig. 2). The median $\mathrm{RMSE_h}$ of the raw forecast rises from 6.3 to 13.1 $\mathrm{m\,s^{-1}}$, and that of WindFuse-0 and FuXi-CFD rises with it. WindSeer does not see the forecast and is insensitive to the corruption (7.8 $\mathrm{m\,s^{-1}}$), but its correlation stays low (0.39): half of the trajectories lie in lee and recirculation zones, which say little about the flow elsewhere. NWP-opt and OI remove much of the error but add little spatial structure, so their $\mathrm{PCC_h}$ stays below 0.5. WindFuse keeps a high correlation (0.71 to 0.65), and its median $\mathrm{RMSE_h}$ grows only from 5.4 to 7.0 $\mathrm{m\,s^{-1}}$. In the large bin NWP-opt reaches the same $\mathrm{RMSE_h}$ as WindFuse, but its correlation is far lower (0.38 against 0.65); at every severity WindFuse correlates 0.24 to 0.27 better. All fitted methods underestimate the speed as the corruption grows, WindFuse less than NWP-opt.

5.2 Perdigão: main result

Figure 3 shows the per-case horizontal errors and Table 1 the case-level medians. The raw forecast has a median $\mathrm{MAE_h}$ of 3.54 $\mathrm{m\,s^{-1}}$, a speed bias of $+1.6\,\mathrm{m\,s^{-1}}$, and little spatial correlation ($\mathrm{PCC_h}$ 0.20, Table 2). The surrogate alone already reduces the error: WindFuse-0 reaches an $\mathrm{MAE_h}$ of 2.42 $\mathrm{m\,s^{-1}}$ and a correlation of 0.77. WindFuse has the lowest error and the highest correlation at every N, with a small negative speed bias, and over half of its gain over WindFuse-0 already comes from a single tower. With one

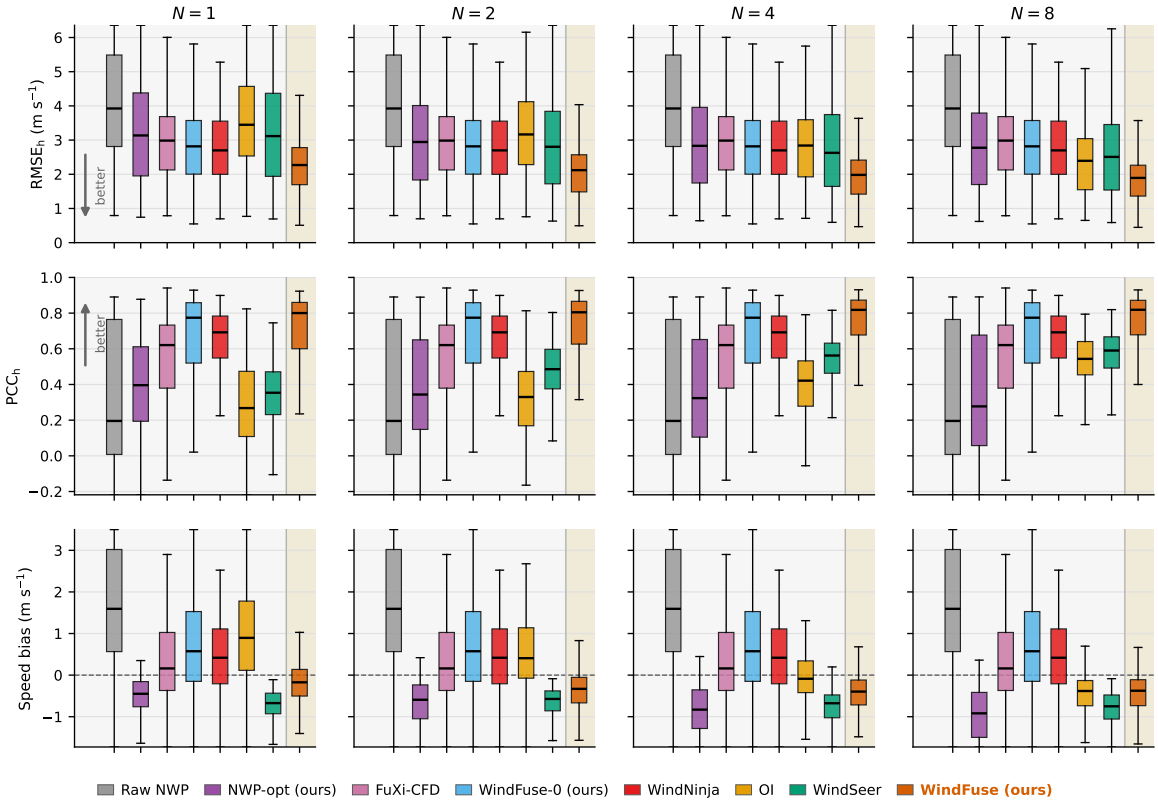


Fig. 3 Perdigão, 200 cases, verified at all withheld sonic anemometers at N=1, 2, 4, and 8 input towers. Rows: horizontal vector RMSE, PCC_h, and speed bias (predicted minus observed). One value per case, averaged over its 30 tower subsets. Observation-free methods do not depend on N.

tower its median MAE_h is 1.98 m s^{-1}, against 3.54 for the raw forecast, 2.58 for WindSeer, which uses the observations alone, 2.26 for WindNinja, and 2.58 for FuXi-CFD; with eight towers it reaches 1.66 m s^{-1}, and only the in-sample OI* is lower (1.47). OI with eight towers stays above WindFuse with one ($RMSE_h$ 2.4 against 2.3, Table 2). Restricted to the 10-m sonics, where WindNinja is designed to operate, the margin is narrower (1.80 against 1.90 m s^{-1}, ESM Table S2) than over the full 2–100 m profile. NWP-opt overcorrects the

Table 1 Perdigão, 200 cases, verified at all withheld sonic anemometers: case-level medians of the horizontal vector MAE (MAE_h) and the speed bias in $m\,s^{-1}$, and the share of cases in which WindFuse has the lower MAE_h (WF better), each with its 95% day-bootstrap interval below. One value per case, averaged over its 30 tower subsets as in Fig. 3. Observation-free methods do not depend on N. Bold: best value per column, smallest magnitude for the bias, in-sample OI* excluded.

	MAE_h		speed bias		WF better	
Method	N=1	N=8	N=1	N=8	N=1	N=8
Raw NWP	3.54		+1.59		100%	100%
	[3.21, 4.19]		[+1.22, +2.14]		[100, 100]	[100, 100]
FuXi-CFD	2.58		**+0.16**		92%	98%
	[2.28, 2.82]		[−0.02, +0.39]		[88, 96]	[95, 100]
WindNinja	2.26		+0.42		78%	90%
	[2.06, 2.52]		[+0.28, +0.62]		[70, 85]	[84, 94]
NWP-opt	2.65	2.40	−0.45	−0.92	86%	96%
	[2.31, 3.11]	[2.04, 2.78]	[−0.53, −0.36]	[−1.11, −0.68]	[80, 92]	[91, 99]
OI* (in-sample)	2.59	1.47	+0.07	−0.46	90%	47%
	[2.28, 2.85]	[1.31, 1.69]	[−0.05, +0.18]	[−0.56, −0.37]	[85, 95]	[39, 54]
WindSeer	2.58	2.16	−0.67	−0.75	84%	84%
	[2.20, 3.05]	[1.82, 2.52]	[−0.82, −0.59]	[−0.85, −0.63]	[78, 91]	[77, 89]
WindFuse	**1.98**	**1.66**	−0.17	−0.38	–	
	[1.83, 2.11]	[1.50, 1.75]	[−0.28, −0.09]	[−0.50, −0.28]		

speed (bias -0.45 to $-0.92\,m\,s^{-1}$), and its correlation falls as towers are added (0.40 to 0.28), since a global rescaling cannot add spatial structure. Direction errors (ESM Table S1, observed speed above $2\,m\,s^{-1}$) tell the same story: the median direction MAE falls from 30.7° (raw forecast) and 25.7° (WindFuse-0) to 17.9° (N=1) and 13.7° (N=8). Among

the operationally applicable methods only WindNinja is comparable (16.5°), and only the in-sample OI* is lower at N=8 (10.3°).

WindFuse has a lower $\mathrm{MAE_h}$ than every operationally applicable method in 78% to 100% of the cases at N=1 and 8 (Table 1), with day-bootstrap intervals above one half and paired Wilcoxon $p < 10^{-16}$ on the per-case values (section 4.7). Against the in-sample OI* it is better in 90% of the cases at N=1 but in 47% $[39, 54]$ at N=8, where OI* has the lower median ($p = 0.04$).

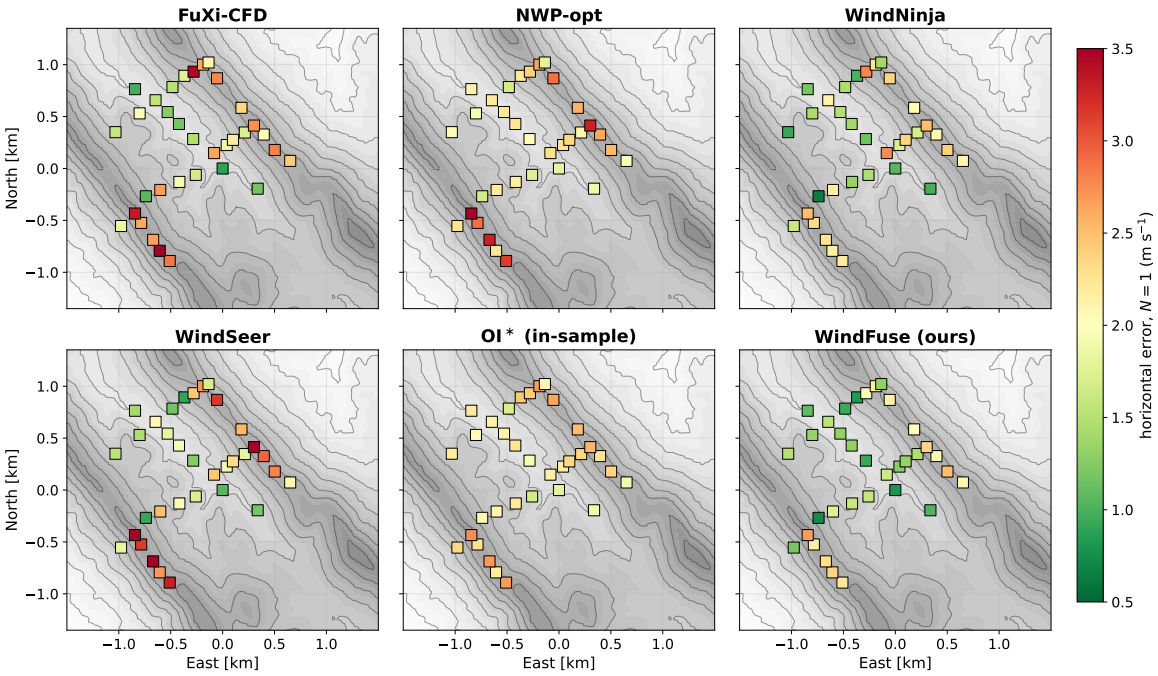


Fig. 4 Perdigão per-tower horizontal vector error at N=1 on the terrain (darker grey is higher). Each square is the median over the 200 cases of the tower's sonics, averaged over the tower subsets in which the tower is withheld. Colours saturate at 3.5 m s^{-1}.

The innovation covariance fitted for OI* decays on a horizontal scale of 2.2 km, com-

parable to the site extent. OI* improves markedly on the operational OI, and its error falls steeply with the number of towers: WindFuse is better at N=1, 2, and 4 ($\mathrm{MAE_h}$ 1.98, 1.83, and 1.72 against 2.59, 2.18, and 1.82 $\mathrm{m\,s^{-1}}$; 90%, 80%, and 64% of the cases). At N=8 OI* has the lower median (1.47 against 1.66 $\mathrm{m\,s^{-1}}$). A poorly configured interpolation is thus not the reason for the gap; even against interpolation tuned on the site's own errors, WindFuse remains competitive without site-specific tuning.

Figure 4 maps, for each tower, its error over all 200 cases (the median over cases) with a single input tower. WindFuse has the lowest median error over the towers (1.6 $\mathrm{m\,s^{-1}}$) and is the best method at 20 of the 34 towers, and its worst tower (2.7 $\mathrm{m\,s^{-1}}$) is the least severe of all methods, whereas WindSeer, NWP-opt, and FuXi-CFD reach 3.5 to 4.1 $\mathrm{m\,s^{-1}}$ on the ridges.

5.3 Recovered corrections

A single correction for the whole campaign cannot replace the per-case fit (Fig. 5, bottom). Applying the median of the 200 fitted corrections at N=8 to every case lowers $\mathrm{RMSE_h}$ from 2.8 $\mathrm{m\,s^{-1}}$ without correction to 2.3 $\mathrm{m\,s^{-1}}$, whereas the case-specific correction improves to 1.9 $\mathrm{m\,s^{-1}}$ with eight towers. Part of the correction is systematic: the speed factor is below one in 86% of the cases, consistent with the representational bias of the pressure-level inflow and with the overestimation of near-surface wind by mesoscale models over complex terrain (section 4.2). The fitted corrections, however, scatter widely around their median (Fig. 5, top), so one (α, φ) fits few cases well. WindFuse thus does

more than remove a constant forecast bias: it corrects each forecast with the observations of that case.

5.4 Dependence on regime

Figure 6 stratifies the case-level scores by time of day and by wind class, which here serves as a proxy for stability (strong winds are near-neutral; the weak-wind class holds the stable nocturnal and convective daytime cases). WindFuse has the lowest $\mathrm{RMSE_h}$ and the

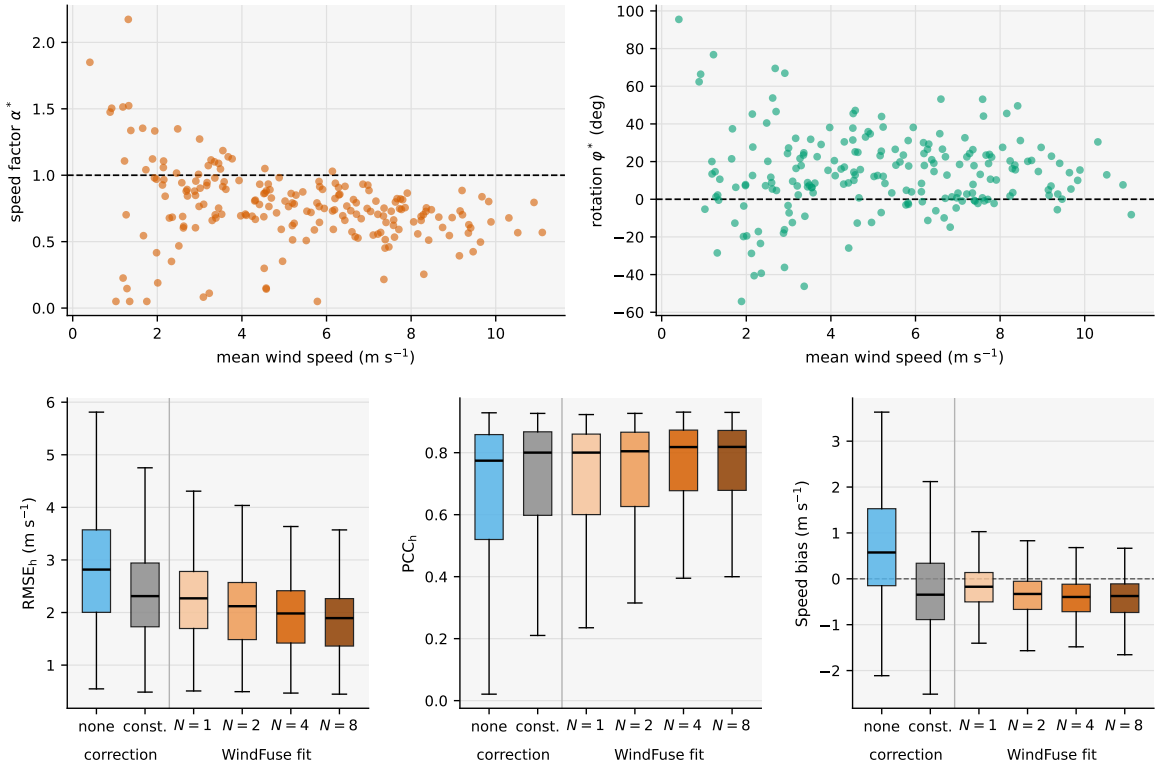


Fig. 5 Fitted corrections on Perdigão. Top: α^* and φ^* of the 200 cases at N=8 against the mean wind speed. Bottom: no correction (WindFuse-0), the single campaign correction (const.), and the WindFuse correction at N=1, 2, 4, and 8, verified at all withheld towers.

highest correlation in every stratum, with a moderate speed bias of -0.2 to $-0.5\,\mathrm{m\,s^{-1}}$. Its margin is largest at midday (1.9 against $2.7\,\mathrm{m\,s^{-1}}$ for OI) and in strong, synoptically driven winds (2.2 against $3.0\,\mathrm{m\,s^{-1}}$ for OI). In these conditions vertical mixing couples the near-surface flow to the forecast winds aloft, which the correction acts on. The margin narrows in weak winds (1.36 against $1.54\,\mathrm{m\,s^{-1}}$ for WindSeer), where the correlation of WindFuse also drops from 0.86 to 0.71. There the near-surface flow is increasingly thermally driven; daytime slope and valley circulations and nocturnal drainage flows lie outside the neutral surrogate. Results for 5-min and 1-h averaging windows are similar (1.95 and $1.85\,\mathrm{m\,s^{-1}}$ for WindFuse).

5.5 Sensitivity to the forecast background

We repeat the 200-case evaluation with the inflow from the CERRA reanalysis (section 4.2) instead of TIGGE (Table 2). CERRA mainly lowers the speed bias of the methods that use no observations: from $+1.6$ to $+1.0\,\mathrm{m\,s^{-1}}$ for the raw forecast and from $+0.6$ to $+0.2\,\mathrm{m\,s^{-1}}$ for WindFuse-0. Their vector error and correlation barely change. With the widened base prior (section 4.7), WindFuse reaches nearly the same error and correlation on either background (1.89 and $1.91\,\mathrm{m\,s^{-1}}$ at N=8) and stays ahead of every baseline. The gain from fusing a few observations is thus largely independent of the forecast background.

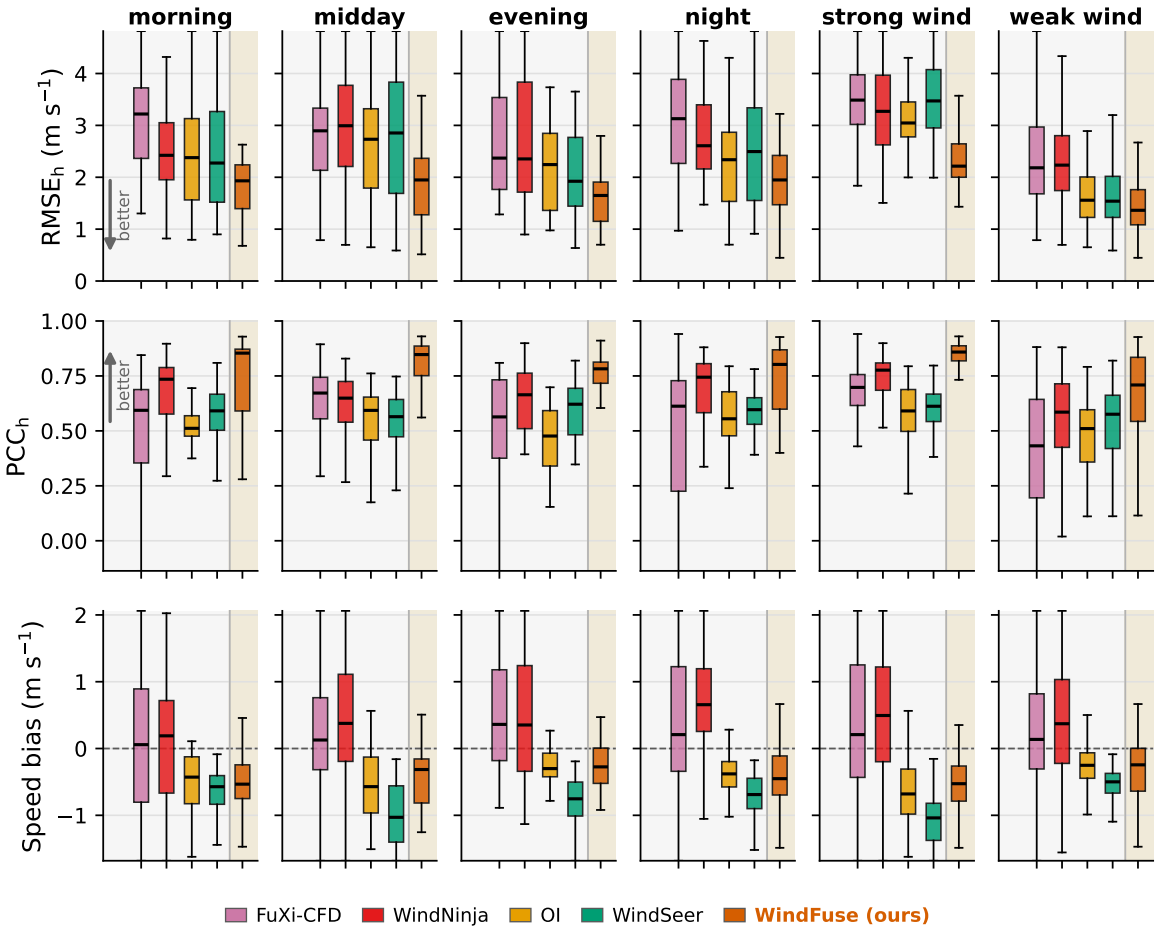


Fig. 6 Perdigão case-level horizontal metrics at N=8 input towers stratified by regime. Columns: time of day of the window start (UTC; at 7.7°W within half an hour of local solar time; morning 05–10, midday 10–17, evening 17–21, night 21–05) and wind class (strong and weak, split at 8 m s^{-1} mean tower wind; 99 and 101 cases). Rows and scoring as in Fig. 3; FuXi-CFD and WindNinja are observation-free.

Table 2 Perdigão median horizontal vector RMSE, PCC_h, and speed bias ($m\,s^{-1}$, all withheld towers) at N=1 and 8 for the TIGGE and CERRA backgrounds (200 cases; CERRA with the widened prior of section 4.7), aggregated as in Table 1. Observation-free methods do not depend on N. Bold: best per column, smallest magnitude for the bias.

	TIGGE						CERRA					
	$RMSE_h$		PCC_h		bias		$RMSE_h$		PCC_h		bias	
	N=1	N=8	N=1	N=8	N=1	N=8	N=1	N=8	N=1	N=8	N=1	N=8
Raw NWP	3.92		0.20		+1.59		3.85		0.15		+0.99	
NWP-opt	3.14	2.78	0.40	0.28	−0.45	−0.92	2.94	2.71	0.38	0.28	−0.21	−0.50
OI	3.45	2.39	0.27	0.54	+0.90	−0.38	3.48	2.58	0.24	0.52	+0.48	−0.66
FuXi-CFD	2.98		0.62		**+0.16**		2.99		0.56		**−0.18**	
WindFuse-0	2.82		0.77		+0.57		2.83		0.72		+0.19	
WindFuse	**2.27**	**1.89**	**0.80**	**0.82**	−0.17	−0.38	**2.33**	**1.91**	**0.78**	**0.82**	−0.40	−0.64

5.6 Transfer of the fitted correction

If (α, φ) captures forecast inflow error rather than a bias of the surrogate, it should also help other models driven by the same forecast, without refitting (Table 3). Applied to the inflow of FuXi-CFD, the WindFuse correction at N=8 lowers the median MAE_h by $0.38\,m\,s^{-1}$ on TIGGE and $0.39\,m\,s^{-1}$ on CERRA and improves 86% of the cases on both. FuXi-CFD then underestimates the speed ($-0.5\,m\,s^{-1}$ on TIGGE). Applied to WindNinja

on TIGGE, the correction lowers the error by a median $0.19\,\mathrm{m\,s^{-1}}$ and improves 74% of the cases. On CERRA it does not help WindNinja, whose 10-m wind is already too slow (speed bias $-0.6\,\mathrm{m\,s^{-1}}$): the reanalysis carries less of the forecast speed bias, so a speed factor below one pushes it further from the truth. The results thus suggest that the correction carries over to other models driven by the same forecast; the CERRA case is less conclusive, because CERRA is a reanalysis rather than a forecast and no higher-resolution forecast was available.

Table 3 Transfer of the WindFuse correction (N=8) to FuXi-CFD and WindNinja on Perdigão, TIGGE and CERRA backgrounds: per-case change of the horizontal vector MAE (median, 95% interval below), share of cases improved, and median speed bias before and after, in $\mathrm{m\,s^{-1}}$, at the sonic anemometers withheld from the N=8 fits. The WindNinja bias before transfer therefore differs by 0.01 from Table 1, which scores all sonic anemometers. WindNinja fails on one CERRA case (199 cases in that row).

	TIGGE forecast			CERRA reanalysis		
Transfer	$\Delta\mathrm{MAE_h}$	improved	bias	$\Delta\mathrm{MAE_h}$	improved	bias
FuXi-CFD, fit (N=8)	−0.38	86%	+0.16→−0.50	−0.39	86%	−0.18→−0.75
	[−0.57, −0.30]			[−0.50, −0.29]		
WindNinja, fit (N=8)	−0.19	74%	+0.43→−0.43	+0.11	34%	−0.56→−1.19
	[−0.28, −0.12]			[+0.04, +0.21]		

5.7 *Airborne observations*

Table 4 Airborne verification: mean $RMAE_{3D}$ and PCC_{3D} per terrain (Gra: Granite Mountain aircraft passes, verified at ground towers; Cha, Got, Obe: alpine sites, leave-one-UAV-out at N=1). Bold: best value per column.

	$RMAE_{3D}$				PCC_{3D}			
Method	Gra	Cha	Got	Obe	Gra	Cha	Got	Obe
Raw NWP	1.01	0.68	0.47	0.65	**0.17**	0.84	0.95	0.92
NWP-opt	1.22	0.44	0.31	0.43	0.12	0.93	**0.99**	0.98
OI	1.07	0.52	0.37	0.56	0.15	0.89	0.97	0.95
FuXi-CFD	0.89	0.89	0.63	0.70	0.14	0.84	0.98	0.92
WindFuse-0	**0.88**	0.60	0.35	0.50	0.13	0.92	0.97	0.96
WindSeer	1.30	0.40	**0.23**	0.31	0.04	0.95	**0.99**	**0.99**
WindFuse	1.04	**0.37**	0.24	**0.30**	0.06	**0.96**	**0.99**	**0.99**

At the alpine sites every method performs well (Table 4), with correlations of 0.84 and higher, because the input UAV samples the flow of interest directly. From a single UAV, WindFuse has the lowest error at two of the three sites and the highest correlation at all three, on par with WindSeer, and it improves on WindFuse-0 at all three alpine sites: the correction does not degrade when the observation comes from a moving platform. At Granite Mountain the ambient flow is weak and the near-surface wind is set by thermally driven slope and valley circulations that neither the forecast nor the aircraft aloft resolve, and no

method reaches a meaningful correlation. With observations that are unrepresentative of the verifying towers, the background term keeps the analysis near the forecast ($RMAE_{3D}$ 1.04 against 1.01 for the raw forecast), whereas the observation-only WindSeer degrades to 1.30. Over all 154 cases, WindFuse has the lowest error (0.42) and the highest correlation (0.84).

5.8 Runtime

A correction typically converges within two of the four outer L-BFGS steps (section 3.6); the remaining steps stop at the tolerance after one evaluation, so a fit takes about 11 forward and backward passes of the surrogate. With a half-precision (FP16) surrogate on an NVIDIA Jetson embedded GPU it takes about 1.4 s for a 64^3 domain (1 km^2) and 3.2 s for a 4 km^2 domain. A WindFuse-0 forward pass takes 30–90 ms, and FuXi-CFD about 10 s.

6. Discussion

Three results support the premise of section 3 that forecast errors over sites a few kilometres across are largely coherent biases. First, the OI* innovation covariance decays over the scale of the site. Second, the correction transfers to two other models driven by the same TIGGE forecast. Third, the speed factor is below one in most cases, consistent with the positive near-surface wind bias of mesoscale forecasts over complex terrain (Xue et al., 2024). The scatter in per-case corrections shows that static bias correction cannot replace daily ob-

servations. WindFuse therefore complements station-level post-processing (Li et al., 2021; Xue et al., 2024), deriving terrain awareness from the flow model rather than a dense observation network (De Pondeca et al., 2011). Where observations are unrepresentative, as over Granite Mountain, the background term keeps the analysis close to the forecast, as required of an assimilation scheme.

Limitations

The surrogate assumes neutral stratification and therefore cannot represent thermally driven flows, including daytime slope and valley circulations and nocturnal drainage and katabatic flows. The dense three-dimensional field is verified directly only on synthetic benchmarks; real-data verification is limited to towers and flight tracks. Dense observational verification, for example against the Perdigão multi-lidar scans (Menke et al., 2019), remains future work. Although the CERRA test shows that the gain persists with a 5.5-km background, a second site and a kilometre-scale forecast would strengthen this conclusion.

Two families of assimilation baselines are absent. Ensemble-covariance schemes (EnKF, EnOI) were not tested because the TIGGE members archived at 50 km do not resolve site-scale structure; their covariance would therefore reduce to a distance-based form similar to OI*. Kilometre-scale variational analysis with an NWP model (e.g., WRFDA) was also not tested because its analysis cannot be evaluated at the tens-of-metres resolution of the dense benchmarks. The raw-forecast and OI rows instead provide comparisons at NWP scale.

7. Conclusions

We presented WindFuse, which fuses numerical weather forecasts with sparse in situ wind observations by fitting a two-parameter speed and direction correction to the forecast inflow through a differentiable CFD surrogate. Its cost function follows the 3D-Var formulation, with a background term informed by ensemble spread, and the correction takes seconds on an NVIDIA Jetson. At Perdigão, a single tower reduces the raw-forecast median error by 44% (from 3.5 to 2.0 $\mathrm{m\,s^{-1}}$) and the WindNinja error by 12%. With one to eight towers, WindFuse outperforms every operationally applicable baseline, with the largest margins under sparse observations and synoptically driven winds. The fitted correction also improves FuXi-CFD and WindNinja when transferred to their inflow under the same TIGGE forecast, suggesting that it captures forecast errors rather than surrogate-specific properties. WindFuse also accommodates moving-platform observations: airborne verification shows the lowest error across all cases, with a fallback to the background when observations are unrepresentative.

Three extensions follow. Richer inflow parameterizations, including vertical shear, spatially varying inflow, and stability, could be regularized under sparse observations by the background term. Temporal persistence of the fitted correction could enable its use between observation windows. Finally, a surrogate trained jointly on forecasts and observations

could fuse both in a single forward pass, eliminating per-case optimization.

Data availability. The processed inputs, the trained networks, the FuXi-CFD-zones cases, and the per-trial results behind every figure and table of this study are available at http://subtdata.felk.cvut.cz/kolibriq/windfuse/. The original sources are the WindSeer CFD dataset, networks, and alpine flight data of the ETH Research Collection (https://doi.org/10.3929/ethz-b-000658323; accessed September–November 2025), the Perdigão 2017 tower data of the NCAR Earth Observing Laboratory (https://doi.org/10.26023/ZDMJ-D1TY-FG14; accessed 28 April 2026), the TIGGE archive at ECMWF and the CERRA pressure-level reanalysis of the Copernicus Climate Change Service (https://doi.org/10.24381/cds.a39ff99f), both accessed June and July 2026, the GFS 0.25° forecasts of the NCAR Research Data Archive (ds084.1, https://doi.org/10.5065/D65D8PWK; accessed June and July 2026), the Copernicus DEM GLO-30 (https://doi.org/10.5270/ESA-c5d3d65) and ESA WorldCover (https://doi.org/10.5281/zenodo.7254221), both accessed June and July 2026, the MATERHORN data of the NCAR Earth Observing Laboratory (https://data.eol.ucar.edu/project/MATERHORN; Twin Otter flight-level data https://doi.org/10.5065/D60000J3; accessed June and July 2026), and the FuXi-CFD dataset and model (https://huggingface.co/linchensen; dataset accessed 18 March 2026, model 19 June 2026).

Code availability. The WindFuse code, with the scripts that produce every figure, table,

and number of this paper from the released data, is available at https://gitlab.fel.cvut.cz/cras/papers/windfuse-aas.

Electronic supplementary material. The ESM gives the construction of the synthetic benchmarks (observation corruption and forecast-like forcing), the OI and OI* configurations, the direction-error and 10-m-sonic verification tables (Tables S1 and S2), and the WindNinja diurnal slope-wind test.

Competing interests. The authors declare that they have no competing interests.

Acknowledgements. This work was co-funded by the Czech Science Foundation under Project 24-12360S, the Grant Agency of the CTU in Prague under Project SGS26/075/OHK3/1T/13, and the European Union under the project Robotics and Advanced Industrial Production (reg. no. CZ.02.01.01/00/22_008/0004590).

References

Achermann, F., N. Lawrance, R. Ranftl, A. Dosovitskiy, J. Chung, and R. Siegwart, 2019: Learning to predict the wind for safe aerial vehicle planning. *IEEE International Conference on Robotics and Automation (ICRA)*, 2311–2317, https://doi.org/10.1109/ICRA.2019.8793547.

Achermann, F., and Coauthors, 2024: Windseer: Real-time volumetric wind prediction over complex terrain aboard a small uncrewed aerial vehicle. *Nature Communications*,

636 **15(1)**, 3507, https://doi.org/10.1038/s41467-024-47778-4.

637 Bannister, R. N., 2017: A review of operational methods of variational and ensemble-
638 variational data assimilation. *Quart. J. Roy. Meteor. Soc.*, **143(703)**, 607–633, https://
639 doi.org/10.1002/qj.2982.

640 Benali, A., A. R. Ervilha, A. C. L. Sá, P. M. Fernandes, R. M. S. Pinto, R. M. Trigo,
641 and J. M. C. Pereira, 2016: Deciphering the impact of uncertainty on the accuracy of
642 large wildfire spread simulations. *Science of the Total Environment*, **569–570**, 73–85,
643 https://doi.org/10.1016/j.scitotenv.2016.06.112.

644 Bougeault, P., Z. Toth, C. Bishop, B. Brown, D. Burridge, and Coauthors, 2010: The
645 THORPEX interactive grand global ensemble (TIGGE). *Bull. Amer. Meteor. Soc.*, **91(8)**,
646 1059–1072, https://doi.org/10.1175/2010BAMS2853.1.

647 De Pondeca, M. S. F. V., and Coauthors, 2011: The real-time mesoscale analysis at
648 NOAA's national centers for environmental prediction: Current status and development.
649 *Wea. Forecasting*, **26(5)**, 593–612, https://doi.org/10.1175/WAF-D-10-05037.1.

650 Dujardin, J., and M. Lehning, 2022: Wind-Topo: downscaling near-surface wind fields to
651 high-resolution topography in highly complex terrain with deep learning. *Quart. J. Roy.*
652 *Meteor. Soc.*, **148**, 1368–1388, https://doi.org/10.1002/qj.4265.

653 European Space Agency, 2021: Copernicus DEM – global and european digital eleva-

tion model, GLO-30. ESA, accessed June and July 2026, https://doi.org/10.5270/ESA-c5d3d65.

Fan, X., X.-Y. Liu, M. Wang, and J.-X. Wang, 2026: Diff-FlowFSI: A GPU-optimized differentiable CFD platform for high-fidelity turbulence and FSI simulations. *Computer Methods in Applied Mechanics and Engineering*, **448**, 118455, https://doi.org/10.1016/j.cma.2025.118455.

Fernando, H. J. S., J. Mann, J. M. L. M. Palma, and Coauthors, 2019: The Perdigão: peering into microscale details of mountain winds. *Bull. Amer. Meteor. Soc.*, **100**, 799–819, https://doi.org/10.1175/BAMS-D-17-0227.1.

Fernando, H. J. S., and Coauthors, 2015: The MATERHORN: Unraveling the intricacies of mountain weather. *Bull. Amer. Meteor. Soc.*, **96(11)**, 1945–1967, https://doi.org/10.1175/BAMS-D-13-00131.1.

Folk, S., J. Melton, B. W. L. Margolis, M. Yim, and V. Kumar, 2024: Learning local urban wind flow fields from range sensing. *IEEE Robotics and Automation Letters*, **9(9)**, 7413–7420, https://doi.org/10.1109/LRA.2024.3426209.

Forthofer, J. M., B. W. Butler, and N. S. Wagenbrenner, 2014: A comparison of three approaches for simulating fine-scale surface winds in support of wildland fire management. Part I. Model formulation and comparison against measurements. *International Journal of Wildland Fire*, **23(7)**, 969–981, https://doi.org/10.1071/WF12089.

673 Frerix, T., D. Kochkov, J. Smith, D. Cremers, M. Brenner, and S. Hoyer, 2021: Variational
674 data assimilation with a learned inverse observation operator. *International Conference*
675 *on Machine Learning (ICML)*, Proceedings of Machine Learning Research, Vol. 139,
676 3449–3458.

677 Gao, H., G. Hu, D. Zhang, W. Jiang, H. Ren, and W. Chen, 2024: Prediction of wind fields
678 in mountains at multiple elevations using deep learning models. *Applied Energy*, **353**,
679 122099, https://doi.org/10.1016/j.apenergy.2023.122099.

680 Geer, A. J., 2021: Learning earth system models from observations: machine learning
681 or data assimilation? *Philosophical Transactions of the Royal Society A*, **379(2194)**,
682 20200089, https://doi.org/10.1098/rsta.2020.0089.

683 Haupt, S. E., and Coauthors, 2019: On bridging a modeling scale gap: Mesoscale to
684 microscale coupling for wind energy. *Bull. Amer. Meteor. Soc.*, **100(12)**, 2533–2549,
685 https://doi.org/10.1175/BAMS-D-18-0033.1.

686 Janjić, T., and Coauthors, 2018: On the representation error in data assimilation. *Quart. J.*
687 *Roy. Meteor. Soc.*, **144**, 1257–1278, https://doi.org/10.1002/qj.3130.

688 Jiménez, P. A., and J. Dudhia, 2012: Improving the representation of resolved and unre-
689 solved topographic effects on surface wind in the WRF model. *J. Appl. Meteor. Climatol.*,
690 **51(2)**, 300–316, https://doi.org/10.1175/JAMC-D-11-084.1.

691 Kalnay, E., 2003: *Atmospheric Modeling, Data Assimilation and Predictability*. Cambridge
692 University Press, 341 pp.

693 Lang, N., W. Jetz, K. Schindler, and J. D. Wegner, 2023: A high-resolution canopy height
694 model of the Earth. *Nature Ecology & Evolution*, **7**, 1778–1789, https://doi.org/10.1038/
695 s41559-023-02206-6.

696 Li, N., L. Ran, D. Shen, and B. Jiao, 2021: An experiment on the prediction of the surface
697 wind speed in Chongli based on the WRF model: Evaluation and calibration. *Adv. Atmos.*
698 *Sci.*, **38(5)**, 845–861, https://doi.org/10.1007/s00376-021-0201-4.

699 Lian, J., and Coauthors, 2024: TerraWind: a deep-learning-based near-surface winds down-
700 scaling model for complex terrain region. *Geophys. Res. Lett.*, **51**, e2024GL112124,
701 https://doi.org/10.1029/2024GL112124.

702 Lin, C., R. Tie, S. Yi, D. Liu, X. Zhong, Z. Hu, and H. Li, 2026: Reconstructing fine-
703 scale 3D wind fields with terrain-informed machine learning. *Nature Communications*,
704 **17**, 3713, https://doi.org/10.1038/s41467-026-70562-5.

705 Liu, N., and Coauthors, 2022: Meshless surface wind speed field reconstruction based on
706 machine learning. *Adv. Atmos. Sci.*, **39(10)**, 1721–1733, https://doi.org/10.1007/s00376-
707 022-1343-8.

708 Lussana, C., O. E. Tveito, and F. Uboldi, 2018: Three-dimensional spatial interpolation of

709 2 m temperature over Norway. *Quart. J. Roy. Meteor. Soc.*, **144(711)**, 344–364, https:
710 //doi.org/10.1002/qj.3208.

711 Maulik, R., and Coauthors, 2022: Efficient high-dimensional variational data assimilation
712 with machine-learned reduced-order models. *Geosci. Model Dev.*, **15**, 3433–3445, https:
713 //doi.org/10.5194/gmd-15-3433-2022.

714 Menke, R., N. Vasiljević, J. Mann, and J. K. Lundquist, 2019: Characterization of flow
715 recirculation zones at the Perdigão site using multi-lidar measurements. *Atmos. Chem.*
716 *Phys.*, **19(4)**, 2713–2723, https://doi.org/10.5194/acp-19-2713-2019.

717 Nasstrom, J. S., G. Sugiyama, R. L. Baskett, S. C. Larsen, and M. M. Bradley, 2007: The
718 National Atmospheric Release Advisory Center modelling and decision-support system
719 for radiological and nuclear emergency preparedness and response. *International Journal*
720 *of Emergency Management*, **4(3)**, 524–550, https://doi.org/10.1504/IJEM.2007.014301.

721 Radünz, W. C., J. M. L. Mattuella, and A. P. Petry, 2020: Wind resource mapping and
722 energy estimation in complex terrain: a framework based on field observations and CFD.
723 *Renewable Energy*, **152**, 494–515, https://doi.org/10.1016/j.renene.2020.01.014.

724 Ridal, M., E. Bazile, P. Le Moigne, R. Randriamampianina, S. Schimanke, U. Andrae,
725 and Coauthors, 2024: CERRA, the Copernicus European Regional ReAnalysis system.
726 *Quart. J. Roy. Meteor. Soc.*, **150(763)**, 3385–3411, https://doi.org/10.1002/qj.4764.

727 Serafin, S., and Coauthors, 2018: Exchange processes in the atmospheric boundary layer
728 over mountainous terrain. *Atmosphere*, **9**, 102, https://doi.org/10.3390/atmos9030102.

729 Sousa, J., C. García-Sánchez, and C. Gorlé, 2018: Improving urban flow predictions
730 through data assimilation. *Building and Environment*, **132**, 282–290, https://doi.org/10.
731 1016/j.buildenv.2018.01.032.

732 Termonia, P., and Coauthors, 2018: The ALADIN system and its canonical model config-
733 urations AROME CY41T1 and ALARO CY40T1. *Geosci. Model Dev.*, **11(1)**, 257–281,
734 https://doi.org/10.5194/gmd-11-257-2018.

735 Uboldi, F., C. Lussana, and M. Salvati, 2008: Three-dimensional spatial interpolation of
736 surface meteorological observations from high-resolution local networks. *Meteor. Appl.*,
737 **15(3)**, 331–345, https://doi.org/10.1002/met.76.

738 Wagenbrenner, N. S., and Coauthors, 2016: Downscaling surface wind predictions from
739 numerical weather prediction models in complex terrain with WindNinja. *Atmos. Chem.*
740 *Phys.*, **16**, 5229–5241, https://doi.org/10.5194/acp-16-5229-2016.

741 Wilks, D. S., 2019: *Statistical Methods in the Atmospheric Sciences*. 4th ed., Elsevier,
742 Amsterdam.

743 Xiao, Y., Q. Jia, H. Fan, P. Fua, R. Jenssen, X. Ma, and W. Xue, 2026: WindINR: Latent-

744 state INR for fast local wind query and correction in complex terrain. arXiv preprint
745 arXiv:2605.09511, https://doi.org/10.48550/arXiv.2605.09511.

746 Xue, W., H. Yu, S. Tang, and W. Huang, 2024: Relationships between terrain features and
747 forecasting errors of surface wind speeds in a mesoscale numerical weather prediction
748 model. *Adv. Atmos. Sci.*, **41(6)**, 1161–1170, https://doi.org/10.1007/s00376-023-3087-5.

749 Zanaga, D., R. Van De Kerchove, D. Daems, W. De Keersmaecker, C. Brockmann,
750 G. Kirches, and Coauthors, 2022: ESA WorldCover 10 m 2021 v200. Zenodo, accessed
751 June and July 2026, https://doi.org/10.5281/zenodo.7254221.

752 Zanetta, F., D. Nerini, M. Buzzi, and S. Moss, 2026: Efficient modeling of subkilometer
753 surface wind with Gaussian processes and neural networks. *Artif. Intell. Earth Syst.*, **5(2)**,
754 https://doi.org/10.1175/AIES-D-24-0047.1.

755 Zängl, G., D. Reinert, P. Rípodas, and M. Baldauf, 2015: The ICON (ICOsahedral
756 Nonhydrostatic) modelling framework of DWD and MPI-M: Description of the non-
757 hydrostatic dynamical core. *Quart. J. Roy. Meteor. Soc.*, **141(687)**, 563–579, https:
758 //doi.org/10.1002/qj.2378.